

The Effect of Exchange Rate Volatility and Labor Market Rigidity on Export Growth

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Abstract

The exchange rate is one of the most volatile macroeconomic price variables. Fluctuations in the exchange rate generate volatility in the profits of firms and reduce incentives for firms to enter the market or expand their capacity. In response to exchange rate volatility, firms can reduce labor costs to mitigate the negative impact by firing workers. However, rigidity in the labor market restricts a firm's ability to implement this adjustment. These restrictions can amplify the negative effects of exchange rate volatility on economic outcomes. This study examines the interaction effects of exchange rate volatility and labor market rigidity on export growth using comprehensive panel data.

To quantify the interaction effects of exchange rate volatility and labor market rigidity, we employ a panel fixed effects econometric model, which is analogous to the difference-in-differences approach. We construct a source-target-industry panel for 135 source countries, a maximum of 183 targets, and 17 industries over a span of 14 years from 2005 to 2018.

Our results imply that in a country where labor market rigidity is one standard deviation higher, a one standard deviation increase in exchange rate volatility reduces export growth by 0.31 percentage points. In a subsample analysis, we find that the interaction between exchange rate volatility and labor market rigidity has a more pronounced negative impact on export growth in Organization for Economic Co-operation and Development (OECD) countries compared to non-OECD countries. We also find that there is little treatment heterogeneity across industries.

Key words: Exchange Rate Volatility, International Trade, Labor Market Rigidity, Exports, Comparative Advantage, Employment Protection Legislation

1 Introduction

The exchange rate is one of the most volatile macroeconomic price variables. For example, the standard deviation of the exchange rate between the UK and the United States dollar averaged 2.3% per month between 2005 and 2018, while the standard deviation of the CPI in the UK was 0.003 (see Table A1). When an exporting firm faces a sudden increase in the exchange rate, one possible response is to lay off workers. This adjustment helps exporting firms cope with the negative effects of an abrupt change in the exchange rate.

However, firms may face difficulties in taking such action when restrictions on worker dismissal are in place. In fact, most countries have employment protection regulations that, to some extent, limit firms' ability to fire workers (OECD, 2020). For example, Colombia, France, and Turkey require an interview with the employee before dismissal, while countries such as the Czech Republic, Poland, Sweden, Finland, Norway, and the Slovak Republic mandate consultation with a third party. Given these constraints, it is important to study how exporting firms respond to both increased exchange rate volatility and labor market regulations.

In previous literature, most studies have focused on evaluating the effect of either exchange rate volatility or labor market rigidity on economic outcomes. Taking labor market rigidity into consideration is important. In the absence of adjustment costs, the profit function is a convex function of the output price. Due to Jensen's inequality, an increase in output price uncertainty leads to higher firm profits. This implies that increased uncertainty in price may result in higher input usage and, consequently, an increase in output. However, in the presence of adjustment costs, this is not necessarily true. Therefore, the effect of uncertainty on investment can be either negative or positive (Calderón et al., 2007; Lee and Shin, 2000). Thus, it is important to examine the combined effect of volatility and adjustment costs (labor market rigidity). In particular, prior studies have not explored how exchange rate volatility interacts with labor market rigidity to influence economic factors such as export growth. This gap in the literature motivates our study, in which we examine the effect of exchange rate volatility and its interaction with labor market rigidity on export growth.

To address this research question, we carry out several data-cleaning and processing

steps. First, we compute exchange rate volatility for currency pairs between 135 source countries and multiple destination countries. This measure captures variations in exchange rate fluctuations across different destination countries within the same exporting country. Additionally, labor market rigidity varies across source countries, even within the same target country. This cross-country variation enables us to estimate the interaction effect of exchange rate volatility while controlling for source country fixed effects, target country fixed effects, industry fixed effects, and time fixed effects.

Next, we construct a disaggregated panel dataset at the source country–target country–industry level, covering a 14-year period from 2005 to 2018. This dataset includes export data from 17 manufacturing industries across up to 183 destination countries. Finally, we analyze the effect of the interaction between exchange rate volatility and labor market rigidity on export growth at the source country–target country pair level.

There are two groups of literature related to this study. The first group includes studies that examine the effect of exchange rate volatility on economic outcomes such as productivity (Aghion et al., 2009; Caglayan and Demir, 2014), firm survival and performance (Baggs et al., 2009), investment (Takagi and Shi, 2011; Atella et al., 2003; Bahmani-Oskooee and Hajilee, 2013), economic growth (Barguelli et al., 2018), and trade flow (Caporale and Doroodian, 1994). In a country-level analysis, Aghion et al. (2009) emphasized the role of financial development in mitigating the negative impact of exchange rate volatility on productivity. Analyzing the top 1,000 public and private firms, Caglayan and Demir (2014) provided evidence of the negative impact of exchange rate uncertainty on productivity in Turkey. According to Baggs et al. (2009), firms exposed to volatile exchange rates may face challenges in maintaining longevity and achieving optimal performance. Takagi and Shi (2011) revealed that exchange rate volatility affects investment decisions, potentially influencing foreign direct investment (FDI) flows. Bahmani-Oskooee and Hajilee (2013) explored how firms' investment decisions are influenced by currency fluctuations, impacting overall economic activity. Barguelli et al. (2018) found that exchange rate volatility negatively affects economic growth and emphasized the need for stable exchange rates to foster sustained economic development. Caporale and Doroodian (1994) suggested that fluctuations in currency

values impact the volume and direction of trade.

The second group of literature examines the effect of labor market rigidity on economic outcomes. For instance, it affects productivity (Alpysbayeva and Vanormelingen, 2022; Bassanini et al., 2009; Kang, 2015), trade (Bista and Sawyer, 2019), technology adoption and output (Caparrós et al., 2013), sales growth and investment (Bai et al., 2020), and employment (Autor et al., 2006). Strict employment protection regulations reduce efficiency at the firm level and, consequently, affect firms' economic performance (Bassanini et al., 2009). In his analysis, particularly for developing countries, Bista and Sawyer (2019) found that labor market rigidity can reduce exports. This reduction is particularly significant because it corresponds to a decrease in the intensive margin of exports, which is crucial for the long-term growth of developing countries. Caparrós et al. (2013) explored how labor market rigidity prevents firms from adopting cleaner technologies, leading to a decline in output and reduced competitiveness. Moreover, in an empirical analysis, Bai et al. (2020) found that employment protection regulations adversely affect firms' investment decisions and, consequently, reduce their growth.

The third group of the related literature examines the effect of output price uncertainty on investment. In the absence of adjustment costs, the profit function is a convex function of the output price. Due to Jensen's inequality, an increase in output price uncertainty will increase the profits of firms. This implies that increased uncertainty in price may lead to higher inputs, and as a result, it may increase output. However, in the presence of adjustment costs, this is not necessarily true. Therefore, the effect of uncertainty on investment can be either negative or positive (Caballero and Pindyck, 1992; Lee and Shin, 2000; Li et al., 2019). Caballero and Pindyck (1992) find that an increase in industry-wide uncertainty by twofold raises the required rate of return on new capital by about 20 percent. In an empirical paper, Li et al. (2019) found that the effect of uncertainty is strongly positive for those firms that have a high labor share, low irreversibility of investment, and low market power.

The studies most closely related to this topic are those by Alexandre et al. (2011, 2017) and Ishise (2019). Ishise (2019) investigated the effect of the interaction between exchange rate volatility and wage rigidity, finding that countries with flexible-wage industries may

have a comparative advantage in the face of exchange rate variability compared to industries with sticky wages. Alexandre et al. (2011) theoretically explored the effects of the interaction between labor market rigidity and exchange rate shocks on employment, focusing solely on Portugal. In an empirical analysis, Alexandre et al. (2017) examined the impact of exchange rate shocks and labor market rigidity on employment growth.

This study extends the literature by investigating the interaction effect of exchange rate volatility and labor market rigidity on export growth, an area that remains largely unexplored. Notably, in economics, exchange rate volatility is conceptually distinct from exchange rate shocks. Additionally, this study contributes by utilizing the comprehensive index of labor market regulation rigidity constructed by Campos et al. (2018), which covers more than 140 countries. This dataset allows us to include 135 developed and developing countries in the empirical analysis, whereas previous studies have covered only a limited number of countries. Furthermore, this study provides subsample empirical results for both OECD and non-OECD countries. Lastly, we find that treatment heterogeneity across industries is relatively minimal.

The purpose of this study is to address these gaps and limitations in the existing literature by conducting a comprehensive empirical analysis with a large dataset. This chapter is organized as follows: Section 2 discusses the data and estimation strategy, exploring exchange rate volatility, labor market rigidity, and the econometric model; Section 3 presents the descriptive statistics and main results; and Section 4 provides the discussion. The study concludes with policy recommendations.

2 Data and Estimation Strategy

2.1 Export Data

This study utilizes panel data on source-target-industry pairs for 135 developed and developing countries over a span of 14 years, from 2005 to 2018¹. We include 17 two-digit manufacturing industries. Export growth is the outcome variable. The two-digit industry-level annual export data from 2005 to 2018 were obtained from the Trade Map. Table A3 (see

¹Table A2 in the Appendix presents the list of sample countries used in the study.

Appendix) presents the list of industries used in the study, along with their corresponding Harmonized Commodity (HS) codes. The inclusion of a large number of countries, facilitated by the labor market rigidity index of Campos et al. (2018), increases the sample size and the precision of our estimates.

2.2 Labor Market Rigidity Index

In each country, labor market regulations are introduced to protect employment. These regulations safeguard the interests of the labor force and are designed to ensure a minimum standard of living for workers. They cover various aspects of employee rights, with one key aspect being protection against unfair dismissal by firms. The strictness of firing regulations varies across countries; in some, firms face no restrictions on terminating employees, while in others, stringent rules apply, such as prior notification, severance pay, third-party agreements, or court approvals. Such strict legislative provisions may discourage firms from hiring employees they cannot easily dismiss later if necessary. Consequently, rigid labor market regulations can hinder the free movement of employees and contribute to labor market rigidity (OECD, 2020).

Several indices have been developed to measure labor market rigidity ². However, most of these indices fail to capture key indicators of rigidity or do not account for all dimensions of labor market rigidity. Additionally, some are limited in scope, covering only a small number of countries. For example, the Employment Protection Legislation (EPL) index applies only to OECD countries and does not reflect other dimensions of labor market rigidity, such as social aspects or wage flexibility.

Furthermore, some indices are constructed based on the specific goals of the organization. For instance, the labor market rigidity index developed by Forteza and Rama (2006) on the World Bank (WB) platform measures labor market rigidity in terms of labor costs. When labor costs increase, output and investment decrease. Thus, the primary purpose of this index

²Labor market rigidity is measured based on factors related to laws on permanent contracts and does not account for non-law factors regarding temporary contracts. Therefore, the labor market rigidity index does not capture the rigidity of temporary contracts that are not included in the law. If temporary contracts increase, employment protection regulations may not influence the rigidity of the labor market, and in this sense, the index has limitations.

is to assess how labor market rigidity affects output and investment, and it is constructed based on factors that influence labor costs. The International Labour Organization (ILO) provides indicators for measuring labor market rigidity; however, it does not have its own index. Most of the indicators specified by the ILO focus on employee rights protection and welfare provisions. However, these indicators do not capture all aspects of firing rigidity.

Botero et al. (2004) developed an index based on four core indicators of employment protection: (1) firing costs of workers, (2) firing procedures, (3) the cost of increasing hours worked, and (4) alternative employment contracts. This index is generated as the average of these four indicators, which, in turn, are derived from sub-indices of various other indicators³. However, this index does not account for the social welfare aspect of employment protection.

Most indices have their own limitations, as discussed above, and do not cover all aspects of employment protection. Therefore, this study uses the Labor Market Legislation Rigidity Index (LAMRIG), constructed by Campos et al. (2018), for the empirical analysis⁴. We selected this index for several reasons. First, it is more recent and comprehensive, covering 140 developed and developing countries, whereas other indices either cover a limited number of countries or represent specific groups, such as only developed countries. Second, this index offers greater precision because it includes time variation, being generated at five-year intervals from 1950–54 to 2000–2004, whereas other indices are time-invariant. Notably, it also covers the period before 1995. Third, this index measures labor market rigidity in real terms, capturing all aspects of employment protection and reflecting key factors contributing to rigidity in the labor market.

Campos et al. (2018) developed this index by incorporating not only the four core indicators introduced by Botero et al. (2004) but also additional important factors related to

³(1) Cost of Firing Workers: This measure quantifies the expenses associated with terminating employees. It considers factors such as the required notice period, severance pay, and other penalties imposed by law or collective agreements. For workers with three years of experience, the index calculates the cost of dismissal. In cases where a worker's termination is deemed unlawful, the index assigns the annual wage of the employee as the cost of firing for the firm. The new wage bill encompasses both the regular wages of the remaining workforce and the cost of firing employees. Additionally, this index assesses firing costs by comparing the new wage bill to the previous one.

⁴Thank you to Nauro F. Campos for providing the dataset for the Labor Market Regulation Index (LAMRIG).

labor markets, drawn from 20 conventions ratified by 150 countries under the International Labour Organization (ILO) since the 1940s. These factors cover different categories. The first category pertains to working conditions, including hours of work, weekly rest periods, and paid leave for employees. The second category encompasses job security measures and regulations on employee termination. The third category addresses employment conditions, considering contracts, wages, and personnel management aspects. The final category includes general regulatory provisions for employees, such as labor codes and employment acts.

This index exhibits considerable variation across countries since 1950, ranging from 0 to 3.5. Higher values indicate more rigid employment protection laws. For instance, countries such as the United States, Japan, Canada, the United Kingdom, and Malaysia had average labor market rigidity values of 0.26, 0.32, 0.51, 0.56, and 0.82, respectively, from 1950–54 to 2000–2004, reflecting more flexible employment protection laws. Meanwhile, Spain, Poland, Slovakia, Mexico, Brazil, Germany, the Netherlands, Estonia, Russia, China, Colombia, and Turkey exhibited labor market rigidity with values around 2 during the same period.

The data on labor market rigidity show a decreasing trend in the mean over time, accompanied by an increase in the standard deviation. Furthermore, employment protection regulations have evolved heterogeneously across countries. For example, some countries have undergone substantial changes in their labor market rigidity index, while others have experienced only minor adjustments.

2.3 Exchange Rate Volatility

Exchange rate volatility refers to the wide range of fluctuations in exchange rates. These fluctuations can create uncertainty and increase the risk associated with business and investment. They can also have considerable adverse effects (Oaikhenan and Aigheyisi, 2015). In the existing empirical literature, three models have been proposed for measuring exchange rate volatility. Most researchers calculate the standard deviation of the monthly exchange rate to measure volatility. The second model measures the z-score of the exchange rate

(Morina et al., 2020). The third model uses the generalized autoregressive conditional heteroskedasticity (GARCH) model, which focuses on predicting errors. This model is particularly important for time series analysis, as it accounts for the fact that the volatility of a time series can change over time (Morina et al., 2020).

The exogenous variation in our identification strategy comes from two sources. First, within each source country, different target countries have different exchange rate volatilities, because we calculated pair exchange rate volatility. Second, even within the same target country, different source countries have different labor market rigidity, which generates exogenous variation in the interaction term, even if two target countries happen to have the same exchange rate volatility. Such exogenous variation allows us to identify the effect of the interaction between exchange rate volatility and labor market rigidity on export growth.

In this empirical analysis, we quantified the pairwise exchange rate volatility. Pairwise exchange rate volatility is computed using the following procedure, incorporating multiple data sources:

Step 1: Exchange rates in terms of the currency of the exporting country

First, to identify the source and target countries in pairs, we utilized the export data of the source country to their counterparts from TradeMap for the time frame 2005–2018. The International Monetary Fund (IMF)’s International Financial Statistics (IFS) database provides the nominal exchange rate of a country per United States dollar for all countries, including both source country j and target country k . This database also provides the inflation rates of countries. We obtained data for the period 2005–2018 for the analysis⁵. First, we converted the exchange rates from nominal exchange rates per United States dollar to nominal exchange rates in terms of the currency of the exporting country j :

$$\text{NER}_{jkm t} = \frac{E_{kmt}}{E_{jmt}} \quad (1)$$

⁵Nominal exchange rates and inflation rates in months for Taiwan are obtained from the website of the National Statistics Republic of China (Taiwan). The inflation rate data for missing periods in the International Monetary Fund (IMF)’s database are filled from the Food and Agriculture Organization (FAO). The annual inflation rate data for Bermuda from 2005 to 2018, Uzbekistan from 2005 to 2010, and Yemen from 2015 to 2018 are obtained from the World Bank. The inflation rate data for some missing periods of Syria and Venezuela are obtained from their national statistics office websites.

In this equation, $NER_{jkm t}$ denotes the nominal exchange rate of target country k in terms of the currency of source country j at month m and year t , where E_{kmt} and E_{jmt} denote the nominal exchange rates in terms of the US dollar for target country k and source country j , respectively.

Step 2: Constructing real exchange rates

we converted the nominal exchange rates in terms of the currency of source country j to the real exchange rates in terms of the currency of source country j using country-specific Consumer Price Index (CPI)⁶.

Accordingly, the pair real exchange rate of source country j to target country k in month m and year t can be written as:

$$\text{Pair RER}_{jkm t} = NER_{jkm t} \times \frac{CPI_{kmt}}{CPI_{jmt}} \quad (2)$$

The real exchange rate of target country k in terms of source country j at month m and year t is denoted by $\text{Pair RER}_{kjm t}$. The CPI of target country k and source country j at month m and year t are denoted by CPI_{kmt} and CPI_{jmt} , respectively.

Step 3: Calculated the volatility of pair real exchange rates

In order to obtain the pair real exchange rate volatility, we calculated the mean of the pair exchange rate level:

$$\text{Pair RER Level}_{jkt} = \frac{1}{12} \sum_m \text{Pair RER}_{jkm t} \quad \text{for each } j, k, t \quad (3)$$

Then calculated the standard deviation of the pair real exchange rates:

$$\sigma_{jkt} = \sqrt{\frac{1}{11} \sum_m (\text{Pairs RER}_{jkm t} - \text{Pair RER Level}_{jkt})^2} \quad \text{for each } j, k, t \quad (4)$$

The $\sigma_{ij t}$ obtained from equation (4) represents the uncertainty in the movement of the

⁶To use more adequate CPI data, we calculated country-specific CPI from the inflation rate (which is the percentage change in the monthly CPI from the previous year) by considering that CPI_{t+1} is a function of $\pi_t, \pi_{t+1}, \dots$ and CPI_t (where π stands for inflation and t denotes the time trend). we also assumed that the initial CPI is equivalent to one.

exchange rate, which implies the volatility in real exchange rate of a pair of source country and target countries.

In the main regression estimation, we controlled for the pair real exchange rate level, the log of the real exchange rate, and the first and second lags of the real exchange rates.

2.4 Estimation Strategy

To quantify the interaction effect of exchange rate volatility and labor market rigidity on export growth, the following econometric model is employed:

$$\begin{aligned} \Delta y_{ijkt} = & \beta_0 + \beta_1 \sigma_{jkt} + \beta_2 LMR_j \times \sigma_{jkt} + \beta_3 Z_{jkt} + \eta_j + \eta_k + \eta_i + \eta_t \\ & + \eta_{jt} + \eta_{kt} + \eta_{it} + \eta_{ji} + \eta_{jk} + \eta_{ik} + \epsilon_{jkt} \end{aligned} \quad (5)$$

$j=1, 2, \dots, 135, i=1, 2, \dots, 17, t=2005, 2006, \dots, 2018$

where i, j, k , and t are the indices of industry, source country, target country to export, and year, respectively. Δy_{ijkt} represents the logarithmic pair export growth at the industry level. σ_{jkt} denotes the pair exchange rate volatility. LMR_j is the time-invariant labor market rigidity of source country j . Z_{jkt} is a set of control variables. Since exchange rate volatility is a key component of the interaction term, we control for it. Other control variables include the pair exchange rate level, the log of the pair exchange rate, the first and second lags of the pair exchange rate, and the pair exchange rate volatility multiplied by gross domestic product (GDP) growth.

The terms η_j, η_k, η_i , and η_t represents the source country fixed effect, target country fixed effect, industry fixed effect and time fixed effect, respectively. $\eta_{jt}, \eta_{kt}, \eta_{it}, \eta_{ji}, \eta_{jk}$, and η_{ik} represents the source-time fixed effect, target-time fixed effect, industry-time fixed effect, source-industry fixed effect, source-target fixed effect and target-industry fixed effect, respectively.

The higher correlation among right-hand side variables, such as interaction of exchange rate volatility and labor market rigidity, pair exchange rate level, log of pairs exchange rate, first lag and second lag of pairs exchange rate, may cause the multicollinearity problem.

However, by exploiting fixed effects regression, this issue is treated appropriately.

Panel fixed effects models can measure an observation repeatedly over time for a pair of source and target countries and for a specific industry. In this model, source fixed effects, target fixed effects, industry fixed effects, and time fixed effects are used to control for source country-level factors, target country-level factors, industry-level factors, and time trend shocks that affect export growth individually. Each fixed effect specification is independent of changes from other fixed effect specifications. For example, different source countries or target countries have different gross domestic products (GDP), populations, legislative frameworks, and institutional policies, all of which can affect export growth. Source country fixed effects and target country fixed effects control for observed and unobserved factors at the country level affecting export growth.

We have several industries, and the availability of many industries may reveal different industry-level characteristics in terms of size, use of technology, access to loans, policy changes, etc. The inclusion of such specifications controls for industry-level factors that affect the export growth of industries heterogeneously.

Time fixed effects are included to control for time trend shocks, such as business cycle fluctuations, over time, regardless of the source country, target country, or industry. For example, a global economic recession would affect export growth in all countries and industries. Time fixed effects control for these time-specific shocks. Moreover, we include source-time fixed effects, target-time fixed effects, and industry-time fixed effects. Country-time fixed effects account for country-level factors that influence export growth over time, regardless of the target country or industry. For example, a change in government policy within a country could impact export growth over time. Target-time fixed effects capture country-level factors that affect export growth independently of other fixed effects, such as changes in GDP or policies that influence the export growth of the source country.

Industry-time fixed effects account for industry-level factors that influence export growth during time-specific shocks, such as fluctuations in trade costs and industry-specific booms or busts. For instance, a change in trade policy could affect the export growth of a particular industry over time.

Additionally, we include country-industry fixed effects to absorb factors influencing export growth relative to variations across countries and industries, independent of time periods and target countries. These fixed effects control for country-industry-specific influences on export growth. Finally, we incorporate target-industry fixed effects to account for factors affecting the export growth of the source country relative to variations in the target country and industry, independent of time-period and source-country variations.

3 Empirical Results

3.1 Descriptive Statistics

This study utilizes the source country–target country–industry panel for 135 source countries, 183 target countries, and 17 industries over a span of 14 years from 2005 to 2018. The panel is constructed from a combination of each pair of source country and target country across 17 industries. For a source country that exports to all 183 target countries for a specific industry, such as electronics, this source country can have 183 pairs at one point in time, such as in 2005. This exporting country can possibly have 43,554 pairs for only one industry. These source-target pairs provide valuable insights. For example, the United States–United Kingdom pair involves exports of electronics from the United States to the United Kingdom, or the Japan–Indonesia pair involves exports of automotive products from Japan to Indonesia. A country for a specific industry has a maximum of 183 pairs if the source country exports to those 183 destinations.

However, the export value is zero for several observations of the source-target pairs. We do not know whether these export values are genuinely zero for several pairs in specific periods. Nonetheless, it is not logical for a country to export to all countries in the world, or to all 183 countries in our sample, across all industries and time periods. A country that exports vehicles may have exports to all 183 countries, but if a country does not produce vehicles and their parts, how can it export to all 183 countries? Therefore, the exports can be zero. Moreover, if there is a disturbance in production, then for specific years, exports can also be zero, even for countries that have bulk production every year in that specific industry. On the other hand, there are countries that do not produce or export products from specific industries, such as Kuwait, Qatar, and Kazakhstan, which do not produce goods from the textile industry. Several countries do not produce or export products from the non-metallic mineral industry, the tobacco industry, or the machinery and equipment industry. This implies missing values for these specific industries in some countries.

Therefore, we follow the dataset construction by Feyrer (2019). Following Feyrer (2019), We construct two datasets: (1) the Two Consecutive Periods Dataset and (2) the Balanced

Panel Dataset. We utilize the Two Consecutive Periods Dataset for the main empirical results. This is an unbalanced dataset that includes pairs with continuous export values for two consecutive periods for a specific industry and a specific source-target pair. Observations with pair export values equal to zero are dropped from the sample. The balanced panel is constructed by dropping the entire panel if the pair export value is zero for even one observation, if there is a gap in periods, or if the pair export data is not available for all 14 years. The second dataset is used for robustness checks. In such data, which contains many observations with zero export values, while utilizing a massive number of observations in the millions and controlling for various fixed effects specifications, the approach of Feyrer (2019) proves to be much more effective for obtaining unbiased and consistent empirical results.

Table 1. Descriptive Statistics

Variable	Mean	Std. Dev.	Min	Max
Dependent Variable				
Export (in millions USD)	75.52	1072.79	0.00	197946.56
Difference of logarithm of Export	0.052	1.269	-13.80	14.62
Independent Variable				
LMR (Labor Market Rigidity)	0.00	1.00	-2.05	1.97
Exchange Rate Volatility	0.00	1.00	-0.27	42.02
LMR $\times$ Exchange Rate Volatility	0.00	0.17	-14.21	13.71
Controls & other Variables				
Pairs Real Exchange Rate Level	0.00	1.00	-0.12	168.72
Log of Real Exchange Rate	-0.45	3.63	-14.40	12.39
First Lag of Real Exchange Rate	-0.45	3.63	-14.40	11.24
Second Lag of Real Exchange Rate	-0.46	3.64	-14.40	11.24
GDP Growth Rate $\times$ Exchange Rate Volatility	0.15	0.69	-3.03	63.59
Industries	17			
Source Countries	135			
N	1,362,920			

Note: Labor market rigidity, real exchange rate volatility and exchange rate level are standardized.

Table 1 shows the descriptive statistics of the key variables used in this study. The large number of pairs across several industries over a span of 14 years enabled us to have sufficient variation in the data and cover a substantial proportion of global manufacturing commodity

exports. The outcome variable is the two-digit industry-level export growth. The mean annual export growth is 0.052, which indicates positive annual export growth, with a standard deviation of 1.26. The main right-hand side variable is the interaction of exchange rate volatility and labor market rigidity. In this unbalanced dataset, the number of observations is 1,362,920.

3.2 Main Results

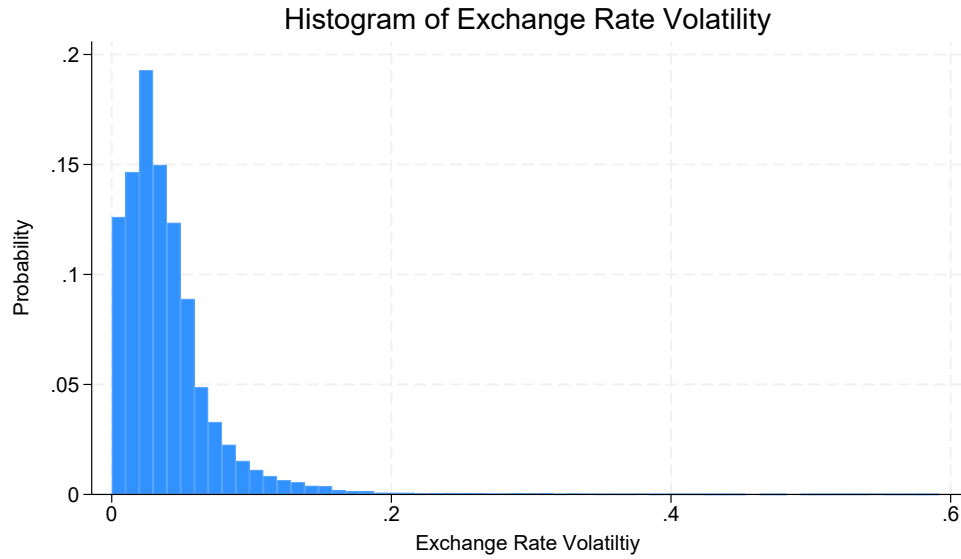
Figure 1 shows the probability distribution of the volatility of the exchange rate. Exchange rate volatility is one of the key components of the explanatory variable, which is calculated as the standard deviation of the exchange rate and is defined as σ_{jkt} in equation (4). A natural question that arises is whether this σ_{jkt} exhibits sufficient variation after controlling for country fixed effects, industry fixed effects, target fixed effects, and time fixed effects.

Figure 2 shows the histogram of exchange rate volatility after controlling for country fixed effects, industry fixed effects, target fixed effects, and time fixed effects. The figure 2 indicates that exchange rate volatility exhibits sufficient variation, even after accounting for these fixed effects. This alleviates concerns that including so many fixed effect specifications in the model might capture all the variation in export growth, leaving no room for the volatility effects of pair exchange rates on export growth.

In this study, a source-industry-target level dataset is constructed to analyze the interaction effects of exchange rate volatility and labor market rigidity on export growth. The regression estimations for the main results are based on the export dataset with two consecutive periods. The sample contains 135 source countries and 17 industries. The dependent variable is export growth, and the main independent variable is the interaction of exchange rate volatility and labor market rigidity.

To quantify the effect of the interaction between exchange rate volatility and labor market rigidity on industrial export growth, we controlled for covariates such as exchange rate level, log of exchange rate, its first and second lags, and the interaction of exchange rate volatility with GDP growth rate. We included source fixed effects, target fixed effects, and time fixed effects in all columns of regression estimations. We also included the source-target fixed

Figure 1: Histogram of exchange rate volatility



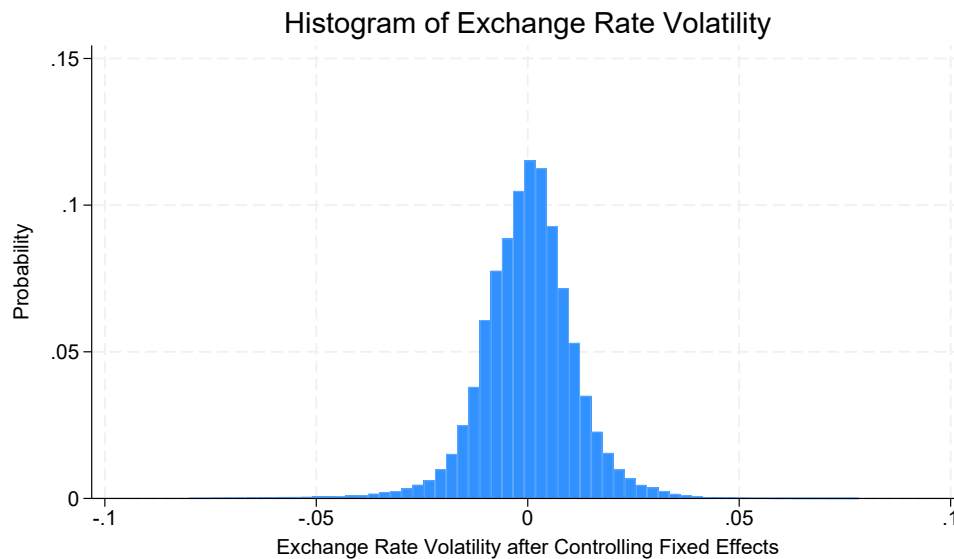
Exchange rate volatility is derived from a panel regression using data from 135 countries for the period 2005-2018.

effect in columns (2)-(8), the source-time fixed effect in columns (2) and (4)-(8), the target-time fixed effect in columns (3)-(8), the industry fixed effect in columns (5)-(8), the source-industry fixed effect in columns (6)-(8), and the industry-time fixed effect in column (8). Column (8) contains all of these fixed effects specifications⁷. The reported standard errors are robust clustered errors in all regression estimations, assuming that error terms are correlated within each source-target.

Table 2 presents the regression estimation of the interaction between exchange rate volatility and labor market rigidity on export growth. The regression estimation in column (8) shows that the estimated coefficient of the interaction between exchange rate volatility and labor market rigidity is negative and statistically significant at the 1 percent level. The estimated coefficient of the interaction term is -0.00311. This indicates that for a country where the rigidity of the labor market is one standard deviation higher, a one standard deviation

⁷Someone might have concerns about financial prices as an exogenous shock affecting export growth. The inclusion of these fixed effects may control for all those country-level factors that affect export growth, including financial crises, etc.

Figure 2: Histogram of Exchange Rate Volatility After Controlling for Fixed Effects Specifications



The above histogram is obtained after regressing the real exchange rate volatility on source fixed effects, industry fixed effects, target fixed effects, time fixed effects, source-time fixed effects, industry-time fixed effects, target-time fixed effects, source-target fixed effects, source-industry fixed effects, target-industry fixed effects, and controls. The controls include the exchange rate level, the log of the real exchange rate, its first lag and second lag, and exchange rate volatility times GDP growth rate. Exchange rate volatility is derived from a panel regression using data from 135 countries for the period 2005–2018.

increase in exchange rate volatility reduces export growth by 0.311 percentage points. The estimated coefficient is also economically significant. The result suggests that the interaction between exchange rate volatility and labor market rigidity reduces annual export growth by 0.311 percentage points. The annual average export growth is 5.2 percentage points, so the economic effect of the interaction term is relatively large. However, the estimated coefficient is smaller relative to the 126.9 percent standard deviation of annual average export growth.

Table 3 provides evidence of the regression estimation of the interaction between exchange rate volatility and labor market rigidity on export growth for OECD countries. In column (8), the coefficient of the interaction term is negative and statistically significant. The sample includes 38 countries. The estimated coefficient remains consistent with the main results in Table 2 and is statistically significant at the 1% level. The estimated coeffi-

Table 2. The effect of the Interaction of Exchange Rate Volatility and Labor Market Rigidity on Export Growth

Dependent Variable	Export Growth							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
LMR $\times$ Exchange Rate Volatility	-0.00322*** (0.000916)	-0.00336*** (0.00121)	-0.00289** (0.00124)	-0.00319*** (0.00119)	-0.00317*** (0.00119)	-0.00311*** (0.00119)	-0.00315*** (0.00120)	-0.00311*** (0.00121)
Exchange Rate Volatility	-0.00449* (0.00230)	-0.00685** (0.00282)	0.00743 (0.0118)	0.00297 (0.0169)	0.00307 (0.0169)	0.00295 (0.0169)	0.00276 (0.0169)	0.00309 (0.0169)
Exchange Rate Level	-0.00288** (0.00130)	-0.00523*** (0.00199)	-0.00535*** (0.00208)	-0.00606*** (0.00193)	-0.00607*** (0.00192)	-0.00607*** (0.00199)	-0.00620*** (0.00201)	-0.00616*** (0.00202)
Observations	1,362,920	1,362,920	1,362,920	1,362,920	1,362,920	1,362,920	1,362,920	1,362,920
R-squared	0.010	0.029	0.030	0.038	0.039	0.041	0.042	0.044
Countries	135	135	135	135	135	135	135	135
Source-Target FE		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Source-Time FE		Yes		Yes	Yes	Yes	Yes	Yes
Target-Time FE			Yes	Yes	Yes	Yes	Yes	Yes
Industry FE					Yes	Yes	Yes	Yes
Source-Industry FE						Yes	Yes	Yes
Target-Industry FE							Yes	Yes
Industry-Year FE								Yes

Note: The two consecutive periods dataset is used for regression estimations. All regressions include source fixed effects, target fixed effects and time fixed effects and controls are log of real exchange rate, first and second lags of log of real exchange rate and exchange rate volatility times GDP growth rate. Clustered robust standard errors are reported, assuming that error terms are correlated within each country $\times$ industry. * indicates significant at 10% level, ** indicates significant at 5% level and *** indicates significant at 1% level.

Table 3. The Effects of the Interaction of Exchange Rate Volatility and Labor Market Rigidity on Export Growth

Dependent Variable	Regression Estimations for OECD Countries							
	Export Growth							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
LMR $\times$ Exchange Rate Volatility	-0.00234*** (0.000827)	-0.00350*** (0.00113)	-0.00290** (0.00118)	-0.00308*** (0.00112)	-0.00306*** (0.00112)	-0.00301*** (0.00112)	-0.00304*** (0.00116)	-0.00299** (0.00116)
Exchange Rate Volatility	-0.00541* (0.00306)	-0.00823** (0.00395)	-0.00896 (0.0190)	-0.0248 (0.0248)	-0.0249 (0.0248)	-0.0246 (0.0248)	-0.0250 (0.0248)	-0.0252 (0.0248)
Exchange Rate Level	-0.00147 (0.00510)	0.00250 (0.00732)	-0.00437 (0.00658)	-0.00551 (0.00702)	-0.00577 (0.00709)	-0.00560 (0.00705)	-0.00614 (0.00706)	-0.00639 (0.00699)
Observations	743,856	743,856	743,856	743,856	743,856	743,856	743,856	743,856
R-squared	0.013	0.020	0.032	0.034	0.035	0.036	0.038	0.040
Countries	38	38	38	38	38	38	38	38
Source-Target FE		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Source-Time FE		Yes		Yes	Yes	Yes	Yes	Yes
Target-Time FE			Yes	Yes	Yes	Yes	Yes	Yes
Industry FE					Yes	Yes	Yes	Yes
Source-Industry FE						Yes	Yes	Yes
Target-Industry FE							Yes	Yes
Industry-Year FE								Yes

Note: The two consecutive periods dataset is used for regression estimations. All regressions include source fixed effects, target fixed effects and time fixed effects and controls are log of real exchange rate, first and second lags of log of real exchange rate and exchange rate volatility times GDP growth rate. Clustered robust standard errors are reported, assuming that error terms are correlated within each country $\times$ industry. * indicates significant at 10% level, ** indicates significant at 5% level and *** indicates significant at 1% level.

cient of the interaction between exchange rate volatility and labor market rigidity is -0.00299. This implies that for a country where labor market rigidity is one standard deviation higher, a one standard deviation increase in exchange rate volatility decreases export growth by 0.299 percentage points. The magnitude of the interaction effect of exchange rate volatility and labor market rigidity is consistent with the estimated coefficient for the entire sample.

Table 4. The Effects of the Interaction of Exchange Rate Volatility and Labor Market Rigidity on Export Growth

Dependent Variable	Regression Estimations for Non-OECD Countries							
	Export Growth							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
LMR × Exchange Rate Volatility	-0.00467* (0.00240)	-0.00242 (0.00294)	-0.00226 (0.00294)	-0.00248 (0.00291)	-0.00247 (0.00292)	-0.00238 (0.00292)	-0.00243 (0.00295)	-0.00239 (0.00294)
Exchange Rate Volatility	-0.00435 (0.00615)	-0.00550 (0.00670)	0.00148 (0.0164)	-0.00851 (0.0268)	-0.00862 (0.0268)	-0.00909 (0.0268)	-0.00878 (0.0269)	-0.00937 (0.0269)
Exchange Rate Level	-0.00310** (0.00130)	-0.00468** (0.00203)	-0.00452** (0.00212)	-0.00496** (0.00193)	-0.00496*** (0.00192)	-0.00496** (0.00199)	-0.00507** (0.00198)	-0.00502** (0.00198)
Observations	619,064	619,064	619,064	619,064	619,064	619,064	619,064	619,064
R-squared	0.008	0.037	0.033	0.046	0.046	0.049	0.051	0.053
Countries	97	97	97	97	97	97	97	97
Source-Target FE		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Source-Time FE		Yes		Yes	Yes	Yes	Yes	Yes
Target-Time FE			Yes	Yes	Yes	Yes	Yes	Yes
Industry FE					Yes	Yes	Yes	Yes
Source-Industry FE						Yes	Yes	Yes
Target-Industry FE							Yes	Yes
Industry-Year FE								Yes

Note: The two consecutive periods dataset is used for regression estimations. All regressions include source fixed effects, target fixed effects and time fixed effects and controls are log of real exchange rate, first and second lags of log of real exchange rate and exchange rate volatility times GDP growth rate. Clustered robust standard errors are reported, assuming that error terms are correlated within each country × industry. * indicates significant at 10% level, ** indicates significant at 5% level and *** indicates significant at 1% level.

Table 4 shows the regression estimates for the interaction effect of exchange rate volatility and labor market rigidity on export growth for non-OECD countries. The number of countries in this sample is 97 with the sample size of 619,064. The estimated coefficient in column (8) is negative but statistically insignificant. The estimated coefficient of the interaction between exchange rate volatility and labor market rigidity is -0.00239. This implies that for a country where labor market rigidity is one standard deviation higher, a one standard deviation increase in exchange rate volatility decreases export growth by 0.239 percentage points.

Tables 3 and 4 presented the subsample regression estimations for OECD and non-OECD

countries. The results suggest that the interaction effect of exchange rate volatility and labor market rigidity on export growth is not significantly different between the two groups of countries. The estimated coefficients are negative; however, they are statistically significant for OECD countries and statistically insignificant for non-OECD countries. The size of the interaction effect is almost the same for both groups of countries. This suggests that exchange rate volatility and labor market rigidity have a relatively more pronounced negative impact on export growth in OECD countries compared to non-OECD countries.

Table 5. The Effects of the Interaction of Exchange Rate Volatility and Labor Market Rigidity on Export Growth

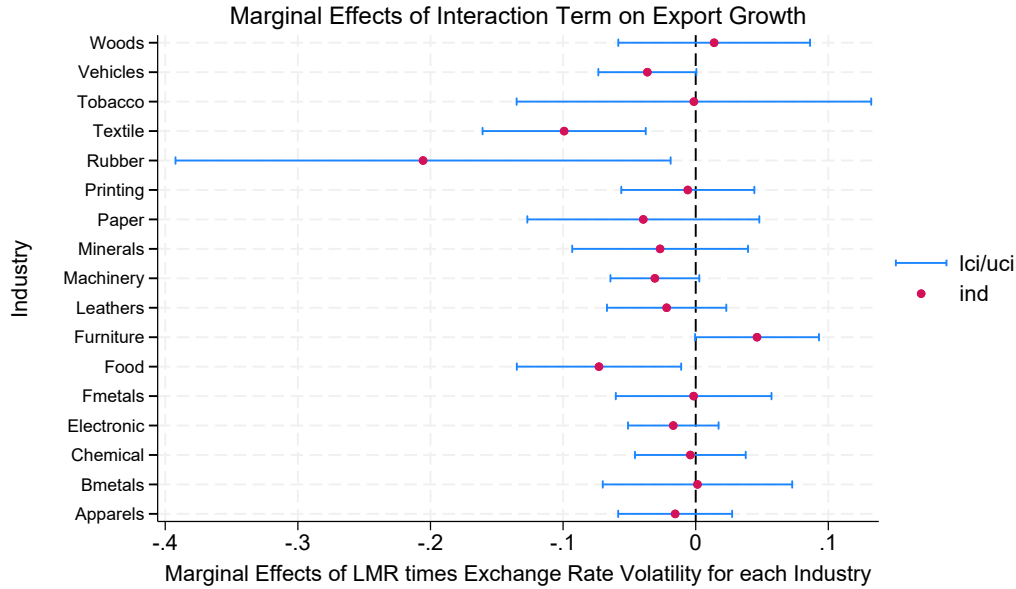
Dependent Variable	Effects of First Lag of Exchange Rate Volatility							
	Export Growth							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
LMR $\times$ Exchange Rate Volatility	-0.00783*** (0.00181)	-0.00822*** (0.00163)	-0.00714*** (0.00173)	-0.00763*** (0.00157)	-0.00761*** (0.00157)	-0.00758*** (0.00157)	-0.00758*** (0.00158)	-0.00754*** (0.00158)
Exchange Rate Volatility	-0.00281 (0.00322)	-0.000745 (0.00345)	0.00291 (0.0122)	0.00799 (0.0172)	0.00804 (0.0172)	0.00869 (0.0172)	0.00887 (0.0172)	0.00880 (0.0172)
LMR $\times$ First Lag of Exchange Rate Volatility	0.00501** (0.00206)	0.00642*** (0.00216)	0.00509** (0.00199)	0.00543*** (0.00195)	0.00544*** (0.00194)	0.00550*** (0.00195)	0.00546*** (0.00194)	0.00546*** (0.00193)
First Lag of Exchange Rate Volatility	-0.00320 (0.00324)	-0.0108** (0.00429)	-0.0319*** (0.0122)	-0.0112 (0.0170)	-0.0111 (0.0170)	-0.0104 (0.0170)	-0.0103 (0.0170)	-0.0103 (0.0170)
Exchange Rate Level	-0.00295** (0.00144)	-0.00459* (0.00240)	-0.00375 (0.00271)	-0.00446* (0.00233)	-0.00446* (0.00231)	-0.00439* (0.00234)	-0.00449* (0.00233)	-0.00448* (0.00233)
Observations	1,166,452	1,166,452	1,166,452	1,166,452	1,166,452	1,166,452	1,166,452	1,166,452
R-squared	0.011	0.031	0.033	0.042	0.042	0.045	0.046	0.048
Countries	135	135	135	135	135	135	135	135
Source-Target FE		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Source-Time FE		Yes		Yes	Yes	Yes	Yes	Yes
Target-Time FE			Yes	Yes	Yes	Yes	Yes	Yes
Industry FE					Yes	Yes	Yes	Yes
Source-Industry FE						Yes	Yes	Yes
Target-Industry FE							Yes	Yes
Industry-Year FE								Yes

Note: The two consecutive periods dataset is used for regression estimations. All regressions include source fixed effects, target fixed effects and time fixed effects and controls are log of real exchange rate, first and second lags of log of real exchange rate and exchange rate volatility times GDP growth rate. Clustered robust standard errors are reported, assuming that error terms are correlated within each country $\times$ industry. * indicates significant at 10% level, ** indicates significant at 5% level and *** indicates significant at 1% level.

Table 5 explores the regression estimation controlling for the interaction of the first lag of exchange rate volatility with labor market rigidity. The interaction of the first lag of exchange rate volatility and labor market rigidity indicates how the previous period's exchange rate volatility affects export growth in the current period. These results show that, after controlling for the previous period's exchange rate volatility and its interaction with labor market rigidity, the main coefficient of the regression remains negative and statistically significant, with the estimated coefficient size becoming larger. The estimated coefficient in column (8) implies that for a country where labor market rigidity is one standard deviation higher, a

one standard deviation increase in exchange rate volatility reduces export growth by 0.75 percentage points. The results are consistent with the main findings.

Figure 3: Marginal Effect of the Interaction of Exchange Rate Volatility and Labor Market Rigidity on Export Growth Across Industries



Notes: The coefficients and confidence intervals in the above-plotted graph are obtained for each industry after regressing export growth on the interaction term (exchange rate volatility and labor market rigidity) interacted with industry code and controls. Controls are pairs exchange rate volatility, pair exchange rate level, and the first and second lags of the logarithmic exchange rate. This analysis controlled for country fixed effects and time fixed effects. The labor market regulation rigidity index developed by Campos et al. (2018) was utilized, and the sample contained 135 countries for the period 2005-2018.

Figure 3 presents the plot of the marginal estimated coefficients for the interaction term of exchange rate volatility and labor market rigidity on export growth across 17 industries. Almost all industries show that export growth is highly sensitive to the interaction of exchange rate volatility and labor market rigidity, except for the wood products and furniture products industries. We find that there is little treatment heterogeneity across industries.

3.3 Robustness Check

In these robustness results, we also constructed a balanced panel where a panel is retained based on the availability of export values greater than zero for all periods from 2005 to 2018. The panel is dropped if export values are zero for a source-target pair; the entire panel is dropped if even one observation has a pair export with a zero value, or if there is a gap in years and pair exports are not available for all periods.

Table 6. The Effects of the Interaction of Exchange Rate Volatility and Labor Market Rigidity on Export Growth								
Dependent Variable	Balanced Panel Estimation with All Periods							
	Export Growth							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
LMR $\times$ Exchange Rate Volatility	-0.00441*** (0.00116)	-0.00500*** (0.00145)	-0.00451*** (0.00142)	-0.00459*** (0.00139)	-0.00459*** (0.00139)	-0.00459*** (0.00139)	-0.00459*** (0.00139)	-0.00452*** (0.00138)
Exchange Rate Volatility	-0.00631** (0.00271)	-0.0102*** (0.00325)	0.0196* (0.0114)	0.0134 (0.0162)	0.0134 (0.0162)	0.0134 (0.0162)	0.0134 (0.0162)	0.0139 (0.0162)
Exchange Rate Level	-0.00264 (0.00172)	-0.00542 (0.00389)	-0.00523 (0.00368)	-0.00626 (0.00400)	-0.00626 (0.00400)	-0.00626 (0.00401)	-0.00626 (0.00401)	-0.00623 (0.00399)
Observations	924,816	924,816	924,816	924,816	924,816	924,816	924,816	924,816
R-squared	0.019	0.036	0.039	0.051	0.051	0.053	0.055	0.057
Countries	113	113	113	113	113	113	113	113
Source-Target FE		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Source-Time FE		Yes		Yes	Yes	Yes	Yes	Yes
Target-Time FE			Yes	Yes	Yes	Yes	Yes	Yes
Industry FE					Yes	Yes	Yes	Yes
Source-Industry FE						Yes	Yes	Yes
Target-Industry FE							Yes	Yes
Industry-Year FE								Yes

Note: All regressions include source fixed effects, target fixed effects and time fixed effects and controls are log of real exchange rate, first and second lags of log of real exchange rate and exchange rate volatility times GDP growth rate. Clustered robust standard errors are reported, assuming that error terms are correlated within each country $\times$ industry. * indicates significant at 10% level, ** indicates significant at 5% level and *** indicates significant at 1% level.

Table 6 indicates the effects of the interaction of exchange rate volatility and labor market rigidity on export growth. The regression estimation in column (8) shows that the estimated coefficient of the interaction of exchange rate volatility and labor market rigidity is negative and statistically significant at the 1% level. The estimated coefficient of the interaction of exchange rate volatility is -0.00452. This indicates that for a country where the rigidity of the labor market is one standard deviation higher, a one standard deviation increase in exchange rate volatility reduces export growth by 0.45 percentage points. The estimated coefficients are economically significant. The result implies that the interaction of exchange rate volatility and labor market rigidity decreases annual export growth by 0.45 percentage

points. The annual average export growth is 5.2 percentage points, so the economic effect of the interaction term is much larger. However, the estimated coefficient is smaller relative to the 126.9% standard deviation of annual average export growth.

The robustness check regression estimations are based on a balanced panel dataset, with a total of 924,816 observations. The regression estimation based on the balanced panel is almost identical to the regression estimations based on the two-consecutive periods unbalanced panel dataset. This indicates that the regression estimates are nearly the same for both the balanced and unbalanced panels, and the results are consistent.

4 Discussion

Previous literature has primarily focused on the effects of exchange rate volatility or labor market rigidity on economic outcomes separately. However, the interaction effect of exchange rate volatility on economic outcomes, which depends on labor market rigidity, has remained largely unexplored. Alexandre et al. (2017) examined the effects of the interaction between exchange rate shocks and labor market rigidity. In contrast, this study investigates the interaction effect of exchange rate volatility and labor market rigidity. Exchange rate volatility is a key component of our main explanatory variable, which is conceptually different from exchange rate shocks. Therefore, this study contributes to the literature by analyzing the effects of exchange rate volatility on export growth in the presence of labor market rigidity.

Our results show that the interaction term between exchange rate volatility and labor market rigidity has a negative coefficient. This finding suggests that the negative effect of exchange rate volatility on export growth is amplified when labor markets are highly rigid. However, the estimated coefficient of exchange rate volatility itself in the same regression is positive.

In theory, the effect of uncertainty on investment can be either negative or positive. Hartman (1972) showed that an increase in output price uncertainty can lead to higher investment by competitive firms. Similarly, Abel (1983), using a different continuous model from Pindyck (1982), demonstrated that even under conditions of known output prices, an increase

in output price uncertainty raises the investment rate. The curvature of the marginal adjustment cost function does not determine the effect of output price uncertainty on investment, indicating that output price uncertainty can have a positive effect on investment.

The literature presents mixed evidence regarding the effects of exchange rate volatility on investment. Li et al. (2019), utilizing the Hartman–Abel model, estimated the effect of exchange rate volatility as a proxy for uncertainty on investment. He showed that the impact of uncertainty is shaped by economic theory. Li et al. (2019) found that uncertainty has a strongly positive effect for firms with a high labor share, low investment irreversibility, and low market power. However, the effect is negative for firms with a low labor share, high investment irreversibility, and high market power. Thus, the impact of uncertainty on investment can be both positive and negative, depending on firm characteristics. In this study, the coefficient of exchange rate volatility is positive, aligning with theoretical expectations.

5 Conclusion

Fluctuations in exchange rates generate volatility in firms' profits and reduce incentives for firms to enter the market or expand their capacity. In response to exchange rate volatility, firms may attempt to reduce labor costs by firing workers. However, rigidity in the labor market restricts their ability to make such adjustments. We argue that these restrictions can amplify the negative effects of exchange rate volatility on economic outcomes.

This study investigated the interaction effect of exchange rate volatility and labor market rigidity on industrial export growth. We constructed a comprehensive panel dataset of 135 source countries and their destinations, covering 17 industries over the period 2005–2018, resulting in a total of 1,362,920 observations.

Our findings suggest that the interaction of exchange rate volatility and labor market rigidity significantly reduces export growth. Specifically, we find that this interaction has a more pronounced negative impact on export growth in Organization for Economic Cooperation and Development (OECD) countries compared to non-OECD countries. We also observe little treatment heterogeneity across industries.

Policy recommendations should move toward a more flexible and balanced labor market.

Flexible labor market regulations are necessary not only to protect employees from labor market risks but also to allow firms to respond to macroeconomic shocks effectively and efficiently.

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Supplemental Information

Table A1. Standard Deviation of Macroeconomic Price Variables

Countries	Variables	Standard Deviation
Japan	Exchange rate Growth	0.023
	CPI Growth	0.003
	Interest Rate Growth	0.000
United Kingdom	Exchange rate Growth	0.023
	CPI Growth	0.003
	Interest Rate Growth	0.002
Germany	Exchange rate Growth	0.022
	CPI Growth	0.003
	Interest Rate Growth	0.002
India	Exchange rate Growth	0.019
	CPI Growth	0.008
	Interest Rate Growth	0.003

Note: The authors calculated the standard deviation from the data.
These are standard deviations of the difference of log of variables.

Table A2. Industries used in this study along with their two-digit HS and ISIC codes

#	Name of Industry	HS 2-digit Code	ISIC 2-digit Code
1	Food and Beverages	22	15
2	Tobacco Products	24	16
3	Textile	53	17
4	Wearing Apparels, fur	61	18
5	Leather, leather products and footwear	42	19
6	Wood Products (excluding furniture)	44	20
7	Paper and Paper Products	48	21
8	Printing and Publishing	49	22
9	Chemicals and Chemical Products	38	24
10	Rubber and Plastics Products	40	25
11	Non-metallic Mineral Products	27	26
12	Basic Metals	83	27
13	Fabricated Metal Products	72	28
14	Machinery and Equipment n.e.c.	84	29
15	Electronic Machinery and Apparatus	85	31
16	Motor Vehicles, Trailers, Semi-Trailers	87	34
17	Furniture; manufacturing n.e.c.	94	36

Note: The ISIC abbreviation stands for International Standard Industrial Classification for all economic activities. The HS stands for Harmonized Commodity description and coding system.

Table A3. List of Countries in the Sample Used for the Empirical Analysis

1 Albania	35 Dominican Rep.	69 Korea, Republic of	103 Romania
2 Algeria	36 Ecuador	70 Kuwait	104 Russian Federation
3 Angola	37 Egypt	71 Kyrgyzstan	105 Rwanda
4 Argentina	38 El Salvador	72 Latvia	106 Saudi Arabia
5 Armenia	39 Estonia	73 Lebanon	107 Senegal
6 Australia	40 Ethiopia	74 Lesotho	108 Seychelles
7 Austria	41 Finland	75 Lithuania	109 Sierra Leone
8 Azerbaijan	42 France	76 Luxembourg	110 Singapore
9 Bahrain	43 Gabon	77 Macedonia	111 Slovakia
10 Bangladesh	44 Gambia	78 Madagascar	112 Slovenia
11 Belarus	45 Georgia	79 Malawi	113 South Africa
12 Belgium	46 Germany	80 Malaysia	114 Spain
13 Benin	47 Ghana	81 Mali	115 Sri Lanka
14 Bolivia	48 Greece	82 Mauritania	116 Suriname
15 Botswana	49 Guatemala	83 Mauritius	117 Sweden
16 Brazil	50 Guinea	84 Mexico	118 Switzerland
17 Bulgaria	51 Guyana	85 Moldova	119 Syrian
18 Burkina Faso	52 Haiti	86 Mongolia	120 Taipei Chinese
19 Burundi	53 Honduras	87 Morocco	121 Tanzania
20 Cambodia	54 Hong Kong	88 Mozambique	122 Thailand
21 Cameroon	55 Hungary	89 Namibia	123 Togo
22 Canada	56 Iceland	90 Nepal	124 Trinidad and Tobago
23 Central African Rep.	57 India	91 Netherlands	125 Tunisia
24 Chad	58 Indonesia	92 New Zealand	126 Turkey
25 Chile	59 Iran	93 Nicaragua	127 United Arab Emirates
26 China	60 Ireland	94 Niger	128 Uganda
27 Colombia	61 Israel	95 Norway	129 Ukraine
28 Comoros	62 Italy	96 Oman	130 United Kingdom
29 Congo Dem Rep.	63 Ivory Coast	97 Pakistan	131 United States
30 Congo Rep.	64 Jamaica	98 Paraguay	132 Uruguay
31 Costa Rica	65 Japan	99 Peru	133 Viet_Nam
32 Croatia	66 Jordan	100 Philippines	134 Yemen
33 Czech Rep.	67 Kazakhstan	101 Poland	135 Zambia
34 Denmark	68 Kenya	102 Portugal	

Note: Countries are selected based on availability of data from different sources.